MODEL IDENTIFICATION AND DATA ANALYSIS (Automation&Control Engineering), A.Y. 2018/2019

Prof. Sergio M. Savaresi, Prof. Simone Formentin - **July 4, 2019**

Surname	**Name**	**University ID Number**	**Signature**
……………………………	……………………………	…………………………………….…	…………………………………….…

==

- Write the solutions (including procedures and intermediate steps) in the blank areas (use the back of the page, if needed)
- The number of pages is 7. Additional papers will not be considered.
- Use of books, lecture notes or any other material is forbidden. Electronics (smartphones, calculators, etc.) are forbidden.
- Clarity, order and precision are strongly considered for the final evaluation.

==

1. [Linear regression]

Given a linear regression problem, consider an initial parameter $\boldsymbol{w}^{(\mathbf{0})} = [0\ 0\ 1]^T$ and a loss function (describing the in-sample error) of the form

$$E_{in}(\boldsymbol{w}) = \frac{1}{2N}\sum_{n=1}^{N}\left(\boldsymbol{w^T x_n} - y_n\right)^2.$$

1.1 Derive the update given from the gradient descent algorithm for the datum $\boldsymbol{x_1} = [1\ 3\ 2]^T, y_1 = 4$ with a step size $\eta = 0.3$.

1.2 What changes in the formula if we perform a batch update using 10 data?

1.3 Discuss the limitations of the gradient descent approach in its original form. Then, explain why the use of the learning rate instead of the step-size and stochastic gradient descent may be helpful to address the above issues.

1.1 The update for a given parameter j for the gradient descent for a single datum is:

$$w_j^{(1)} = w_j^{(0)} - \eta\frac{\partial E_{in}(\boldsymbol{w})}{w_j} = w_j^{(0)} - \eta\left(\boldsymbol{w}^{(\mathbf{0})\boldsymbol{T}}\boldsymbol{x_1} - y_1\right)x_{1j}$$

Thus we have:

$$w_1^{(1)} = 0\ - 0.3(2-4)*1\ =\ 0.6$$
$$w_2^{(1)} = 0\ - 0.3(2-4)*3\ =\ 1.8$$
$$w_3^{(1)} = 1\ - 0.3(2-4)*2\ =\ 2.2$$

And the final coefficient is $\boldsymbol{w}^{(\mathbf{1})} = [0.6\quad 1.8\quad 2.2]^T$.

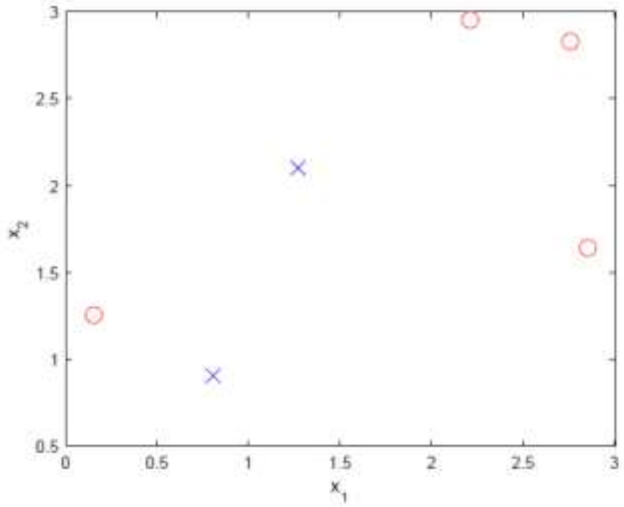
1.2 In the case we want to consider a set of 10 samples, we could use the vector formulation:

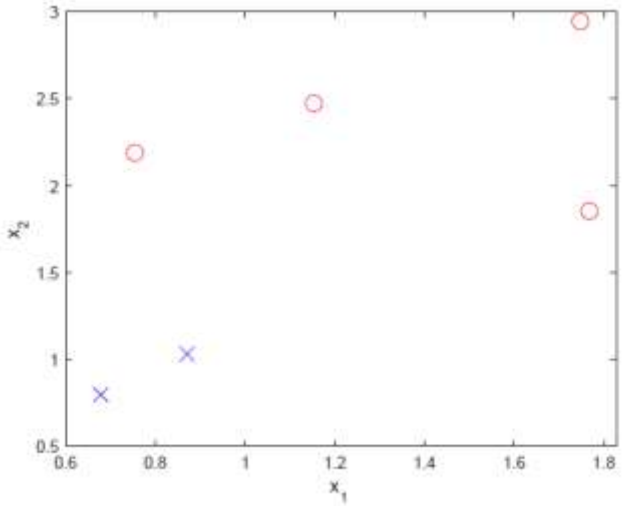
$$\boldsymbol{w}^{(1)} = \boldsymbol{w}^{(0)} - \eta\frac{\partial E_{in}(\boldsymbol{w})}{\boldsymbol{w}} = \boldsymbol{w}^{(0)} - \frac{\eta}{10}\sum_{n=1}^{10}\left(\boldsymbol{w}^{(\mathbf{0})\boldsymbol{T}}\boldsymbol{x_n} - y_n\right)\boldsymbol{x_n}$$

1.3 See Chapter 6 of the lecture notes for the details about the gradient descent algorithm.

2. [Linear classification]

Consider two different datasets (see the left and right figures below) and the PLA algorithm to train a perceptron:





2.1 Describe the perceptron model.

2.2 Under which assumptions the PLA is guaranteed to converge? Provide a formal proof of convergence in such conditions.

2.3 Does the learning procedure terminates in the two cases illustrated in the figures?

2.4 If so, can we say how many steps we require to reach convergence?

2.1 See Chapter 3 of the lecture notes.

2.2 The perceptron learning algorithm is guaranteed to converge in the case it exists a linear separation surface. In such a case, we are able to reduce the classification error to zero, otherwise the optimization procedure does not stop. See Chapter 3 of the lecture notes for the mathematical proof of convergence.

2.3 For the reasons discussed above, in the first case (left) we are sure the algorithm does not converge, while in the second case (right) the online procedure will eventually converge.

2.4 We do not have any assurance about the convergence rate, since it depends on the starting point for the parameter and on the ordering of the points we consider for training.

3. [Neural Networks]

Draw the scheme of a multi-layer feedforward neural network and explain how it overcomes the modeling limits of a simple perceptron.

See Chapter 9 of the lecture notes.

4. **[Kalman filtering]**

Consider the following system in state-space form:

$$\begin{cases} x_1(t+1) = 3x_1(t) + w(t) \\ x_2(t+1) = 3x_1(t) + w(t) \\ y(t) = x_1(t) - \alpha x_2 + v(t) \end{cases}$$

where α is a real number, $w \sim WN(0,1)$ and $v \sim WN(0,1)$ are uncorrelated. Let $\hat{x}(t+1|t)$ be the 1-step ahead Kalman predictor for the above system.

4.1 Explain why, when $\alpha = 0$, the steady-state variance of the 1-step ahead prediction error is limited for any t.

4.2 Explain why, when $\alpha = 1$, the steady-state variance of the 1-step ahead prediction error diverges with t.

4.3 Draw the block scheme of the 1-step ahead Kalman predictor in the latter case.

4.1 When $\alpha = 0$, the system can be rewritten as the composition of

$$\begin{cases} x_1(t+1) = 3x_1(t) + w(t) \\ y(t) = x_1(t) + v(t) \end{cases}$$

which is reachable and observable (thus the prediction error variance is limited for the second Theorem of convergence) and

$$x_2(t+1) = 3x_1(t) + w(t)$$

which describes the dynamics of x_2 (equal to that of x_1). It follows that both the states can be predicted with limited variance.

4.2 When $\alpha = 1$, the output becomes $y(t) = x_1(t) - \alpha x_2 + v(t) = v(t)$, i.e., a white noise, thus it is useless to adjust the estimate the state. Moreover, since the dynamics is unstable, the real trajectory of x_1 (and x_2) may diverge infinitely far away from the estimate.

4.3 See Chapter 16 of the lecture notes.

5. [ARMAX identification]

The following impulse response is measured

t	0	1	2	3
$u(t)$	1	0	0	0
$y(t)$	-1	2	-2	$1/2$

Consider the class of ARX: $M: y(t) = ay(t-1) + au(t-1) + \xi(t)$ $\xi(t) \sim WN(0, \lambda^2)$

a) Consider $u(t) = 0$. Find the range of parameter a such that the model is stationary

b) Identify the value a^* of parameter a which minimizes $J_3(a) = \frac{1}{3}\sum_{i=1}^{3}\left(y(t) - \hat{y}(t|t-1)\right)^2$

c) Give the best estimation of λ^2

$y(t) = \underbrace{\frac{1}{1-az^{-1}}}_{W(z)}\xi(t)$

The process is stationary if W(z) is asymptotically stable.

$1 - az^{-1} = 0$ $z = a$ W(z) is A.S. if and only if $-1 < a < 1$

I. **Predictor of model class**

$M: y(t) = ay(t-1) + au(t-1) + \xi(t)$

$\hat{y}(t|t-1) = ay(t-1) + au(t-1)$

II. **Prediction error**

$\varepsilon(t) = y(t) - \hat{y}(t|t-1) = y(t) - ay(t-1) + au(t-1)$

t	0	1	2	3
u(t)	1	0	0	0
y(t)	-1	2	1/2	-2
ε(t)	/	2	0.5-2a	-2-0.5a

III. **Cost function**

$J_3(a) = \frac{1}{3}\left[(2)^2 + \left(\frac{1}{2} - 2a\right)^2 + \left(-2 - \frac{1}{2}a\right)^2\right]$

$J_3(a) = \frac{1}{3}\left[4 + \frac{1}{4} + 4a^2 - 2a + 4 + \frac{1}{4}a^2 + 2a\right] = \frac{1}{3}\left[\frac{33}{4} + \frac{17}{4}a^2\right]$

IV. **Minimization**

$\frac{\partial J_3(a)}{\partial a} = 0$ $\frac{1}{3}\left[\frac{17}{2}a\right] = 0$ $a^* = 0$

$\lambda^2 = \bar{J}(a^*) = \frac{11}{4}$

6. **[analysis of stochastic processes].** Consider the stochastic process generated according to the following equation:

$y(t) = e(t-1) - e(t-4)$ where $e \sim WN(1,1)$

Is the process $y(t)$ stationary? Why?

Compute the mean value and the variance of the process.

Compute the covariance function $\gamma_y(\tau)$ (with $\tau = 0, \pm 1, \pm 2, \ldots$) of process $y(t)$.

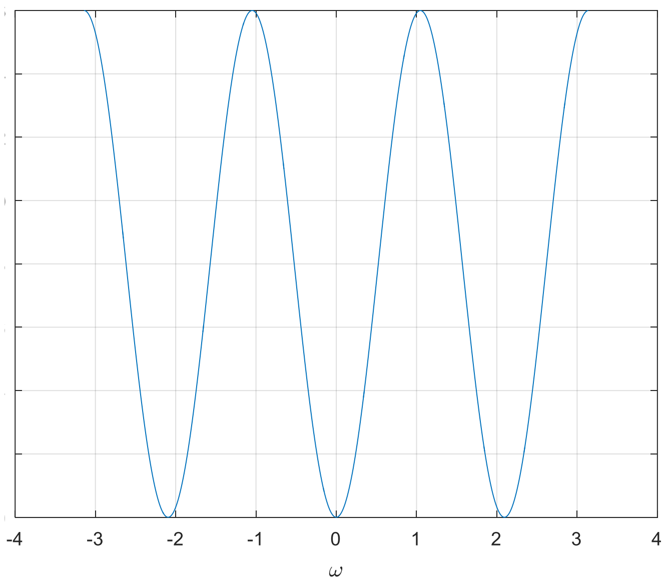
Compute the spectrum of the process y(t) and depict its qualitative plot.

The process is stationary, since it is of MA type.

$\bar{y} = \mathbb{E}[y(t)] = \mathbb{E}[e(t-1)] - \mathbb{E}[e(t-4)] = 0$

$\gamma_y(0) = \mathrm{var}\big(y(t)\big) = 2$, $\gamma_y(\pm 1) = \gamma_y(\pm 2) = 0$, $\gamma_y(\pm 3) = -1$, $\gamma_y(\pm \tau) = 0$ for all $\tau \geq 4$.

The spectrum of $y(t)$ is $\Gamma_y(\omega) = 2(1 - \cos(3\omega))$. Its plot is



7. **[frequency-domain identification]** Describe the type of experiments that must be done on the system in a frequency-domain identification method. Moreover, explain the pre-processing method applied to the experimental data in such methods.

See notes class-notes-14-12-2018 (pp.6-18)